



MUHAMAD RAYHAN FITRA NUGRAHA

TEKNIK KOMPUTER JARINGAN

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EDUCATION

SMP NEGERI 4 TUBAN
2020-2023

SMK NEGERI 1 TUBAN
TEKNIK JARINGAN KOMPUTER

PROYEK ATAU INOVASI

- Developed a smart pest repellent system utilizing the **FOMO (Faster Objects, More Objects)** algorithm for high-speed pest detection on edge devices. This project was **awarded 2nd Place in the National Science Paper Competition at Universitas Airlangga**.
- Developed an early plant disease detection system using a hybrid architecture of **YOLOv8n** for object localization and **EfficientNet-B0** for accurate disease classification. This innovation was **awarded 2nd Place in a National Science Paper Competition**.
- Designed a **YOLOv8n-based** automated pest monitoring system. **3rd Runner Up, ITS**.

PROFILE

A Computer and Network Engineering graduate with specialized expertise in Microcontroller Programming (ESP32), IoT solutions, and Renewable Energy systems. I am a National-level Science Paper Competition winner with a proven track record of creating technological innovations. Beyond development, I possess strong technical skills in network configuration, OS installation, and hardware maintenance, having successfully serviced over 100 PC and laptop units during my internship.

ACHIEVEMENTS

- **2nd Place Winner, LKTI 2026**
2nd Place Winner, National Science Paper Competition ITS, 2026.
- **2nd Place Winner, IT 2025**
2nd Place Winner, National Technology Innovation Competition (UNAIR), 2025.
- **3rd Runner Up, LKTI 2025**
3rd Runner Up, National Instrumentation Competition (INSKILL) - 2025.
- **TRANING INSTUMENTASI ITS**
Technical Instrumentation Training - Hosted by the Department of Instrumentation Engineering, ITS (December 14, 15, and 18, 2025).

INTERNSHIP EXPERIENCE

- **Internship PLTU TANJUNG AWAR-AWAR**
 - Performed system re-installations (Windows/Linux) for 100+ PC and laptop units as part of an IT infrastructure upgrade project.
 - Executed SSD upgrades and installations across multiple devices to significantly enhance system performance and processing speeds.
 - Managed critical data backup and restoration processes to ensure zero data loss during system transitions and installations.
 - Diagnosed and resolved diverse hardware and software malfunctions to maintain hardware reliability.
 - Partnered with the internal IT team to ensure all installations met industry standards and project deadlines.

BOOTCAMP

- **Industrial Robotics Bootcamp - PTC Surabaya, East Java (November 8, 2024)**.
- Completed intensive training focused on the fundamentals of industrial robotic systems and automation.
- Gained hands-on experience in robotic programming and operational workflows within an industrial context.
- Explored the integration of sensors and controllers to optimize robotic performance.